Causal Inference and Machine Learning

In Economics, Social, and Health Sciences

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What This Chapter Covers

- Differences between **prediction** and **estimation**
- How ML is used across economics, health, and social sciences
- Key modeling paradigms: statistical vs. ML, parametric vs. nonparametric, predictive vs. causal
- Introduction to **bias-variance tradeoff** and **simulation**

A Parable: The Blind Men and the Elephant

Different researchers see different facets of data and modeling: - Econometricians: theory-driven, causality-focused - Statisticians: optimality, inference - Computer scientists: algorithmic performance

Prediction vs Estimation

Prediction: Focused on accurate forecasts
Estimation: Targeting model parameters, often causal
Prediction: What will happen?
Estimation: Why and by how much?

Examples: - Will a patient be re-admitted? - Does the new policy reduce crime?

R Code: Predicting with Regression

```r
# Example: Predict income using education and age
lm_model <- lm(income ~ education + age, data = survey_data)
predicted_income <- predict(lm_model, newdata = test_data)
```

Output would show predicted values for a new sample.

ML Applications in Economics & Social Sciences

- Heterogeneous treatment effects (e.g., causal forests)
- High-dimensional controls (e.g., LASSO, Post-LASSO)
- Model selection and regularization
- Policy evaluation

```r
library(glmnet)
x <- model.matrix(y ~ ., data = data)[, -1]
y <- data$y
cv_fit <- cv.glmnet(x, y, alpha = 1)
plot(cv_fit)
```

Key Takeaways

- ML is not just prediction—it can support causal inference
- Know when to use prediction vs. estimation
- This course/book will balance both worlds
- Simulations and exercises will help reinforce these ideas